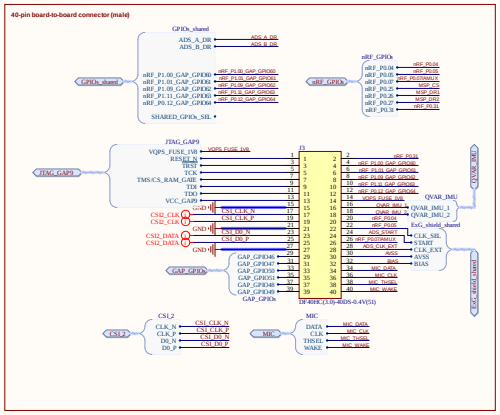
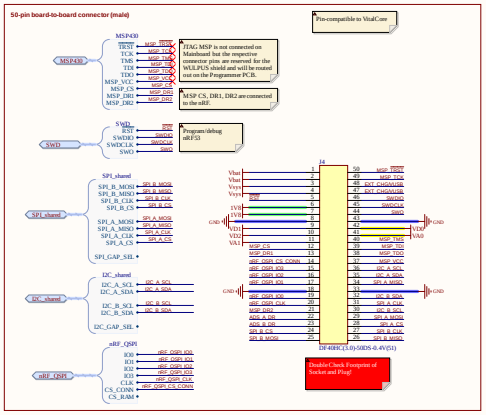
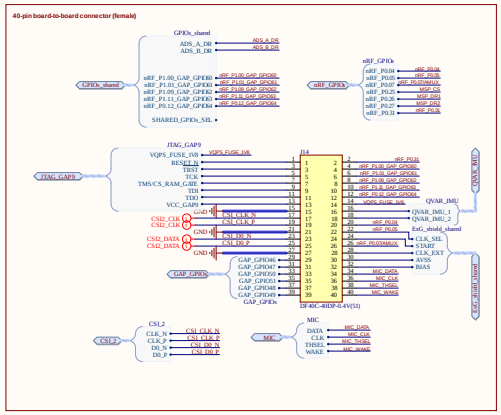
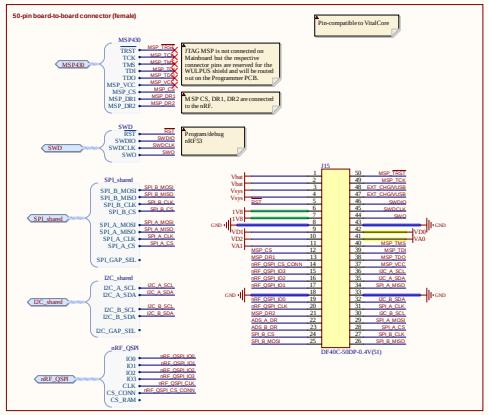
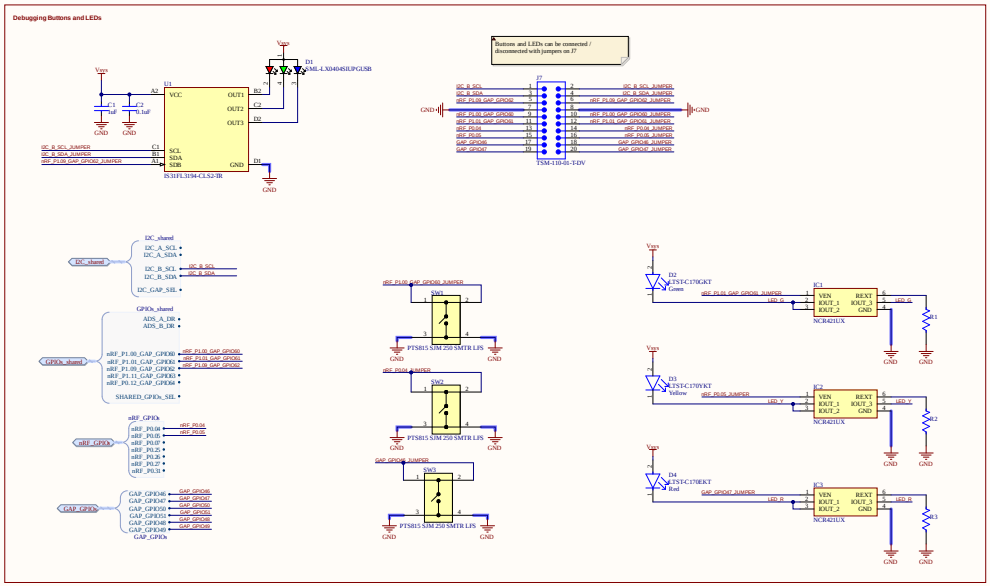
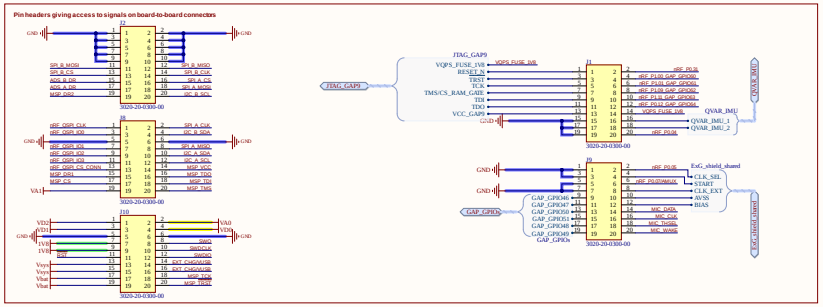


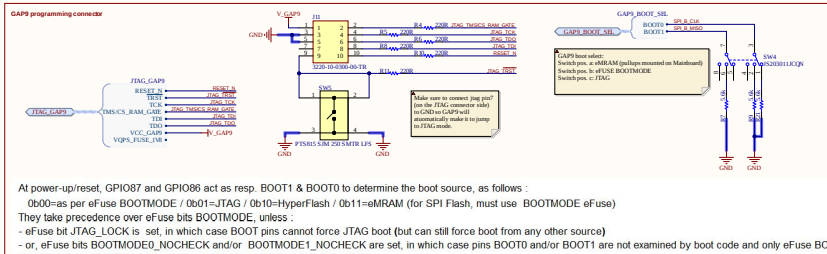












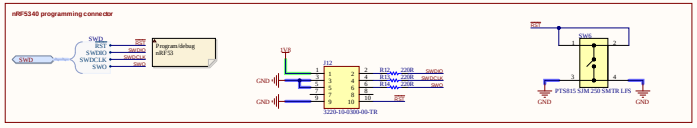
At power-up/reset, GPIO87 and GPIO86 act as resp. BOOT1 & BOOT0 to determine the boot source, as follows :

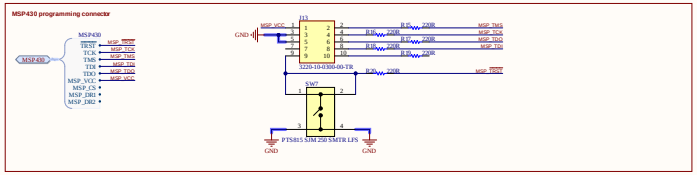
0b00=as per eFuse BOOTMODE / 0b01=JTAG / 0b10=HyperFlash / 0b11=eMRAM (for SPI Flash, must use BOOTMODE eFuse)

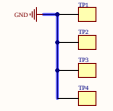
They take precedence over eFuse bits BOOTMODE, unless :

- eFuse bit JTAG_LOCK is set, in which case BOOT pins cannot force JTAG boot (but can still force boot from any other source)
- or, eFuse bits BOOTMODE0_NOCHECK and/or BOOTMODE1_NOCHECK are set, in which case pins BOOT0 and/or BOOT1 are not examined by boot code and only eFuse BOOTMODE settings are taken into account

After boot, these 2 pins revert to their standard role and can be ued as GPIOs





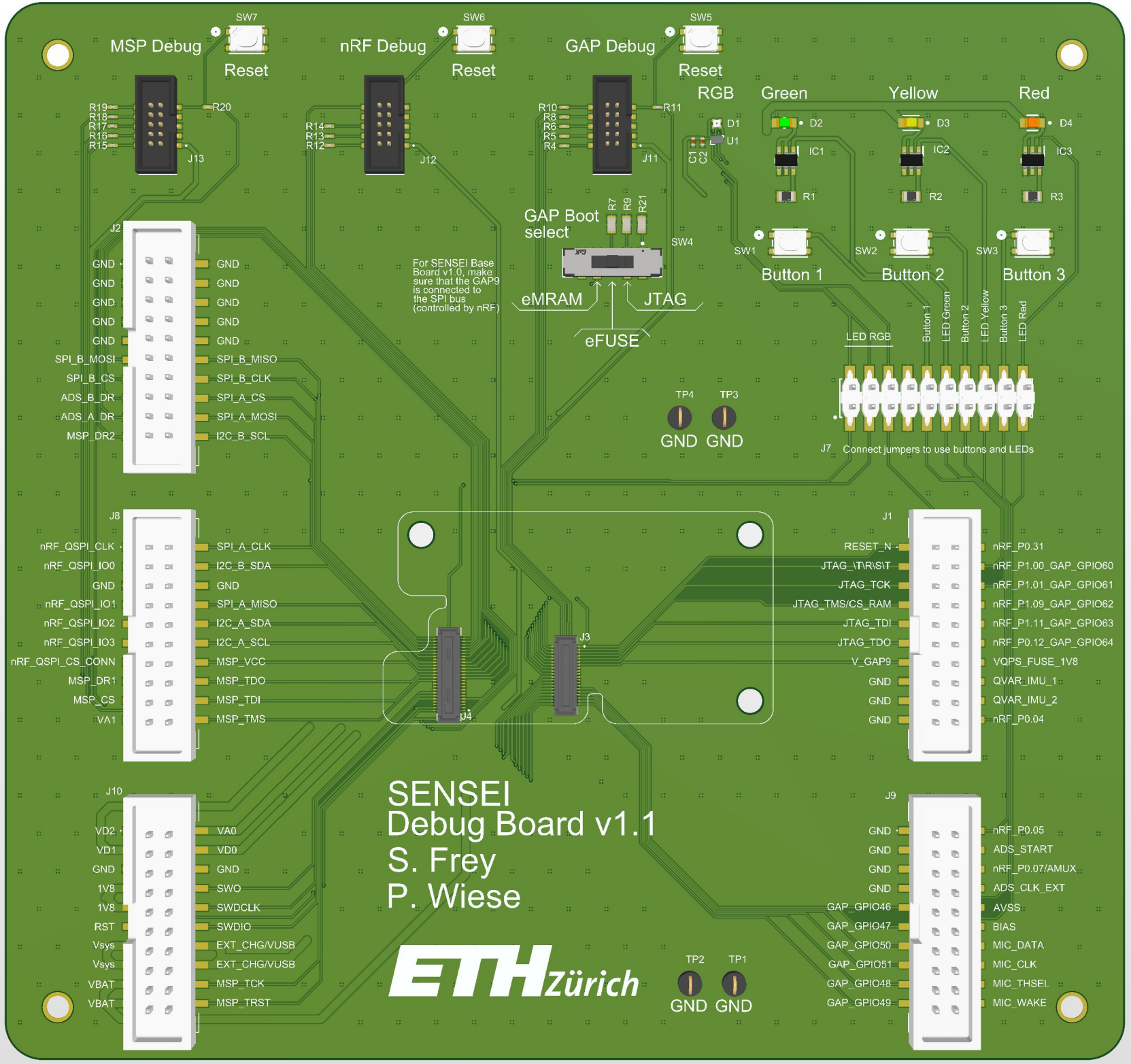


ETH
Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

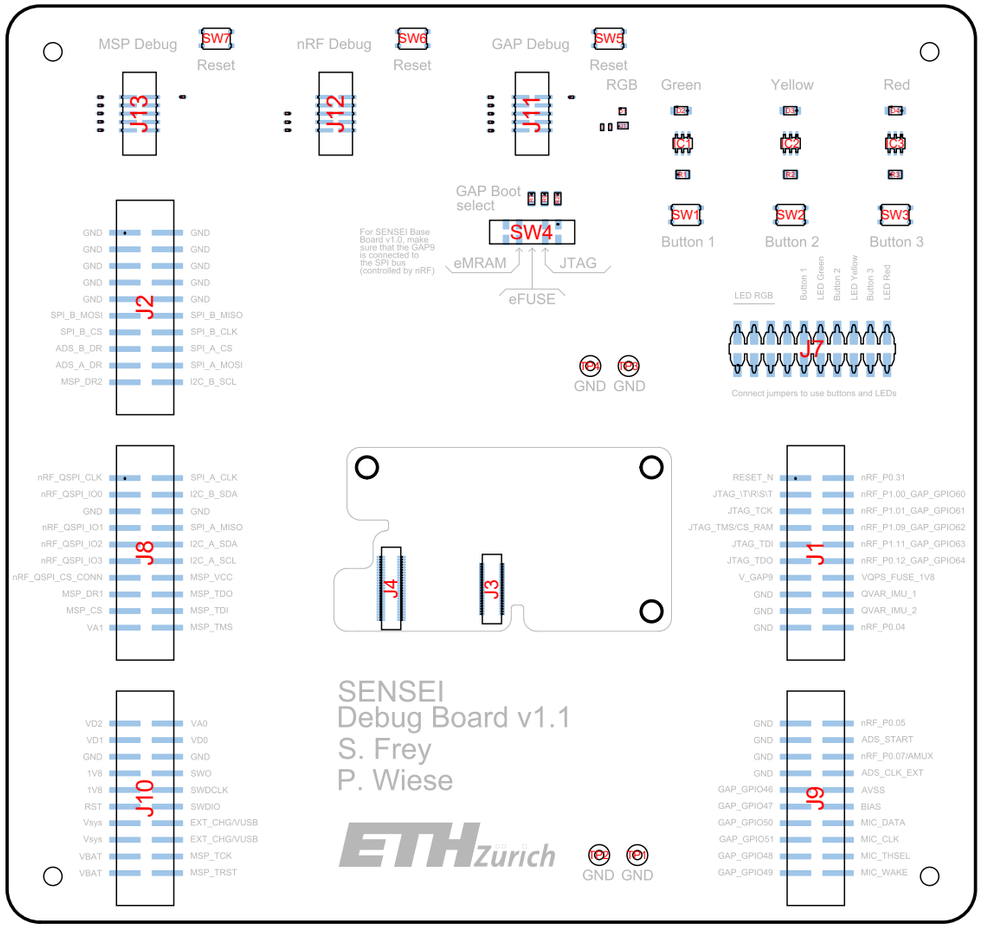
Drawing title:

SENSEI Debug Board

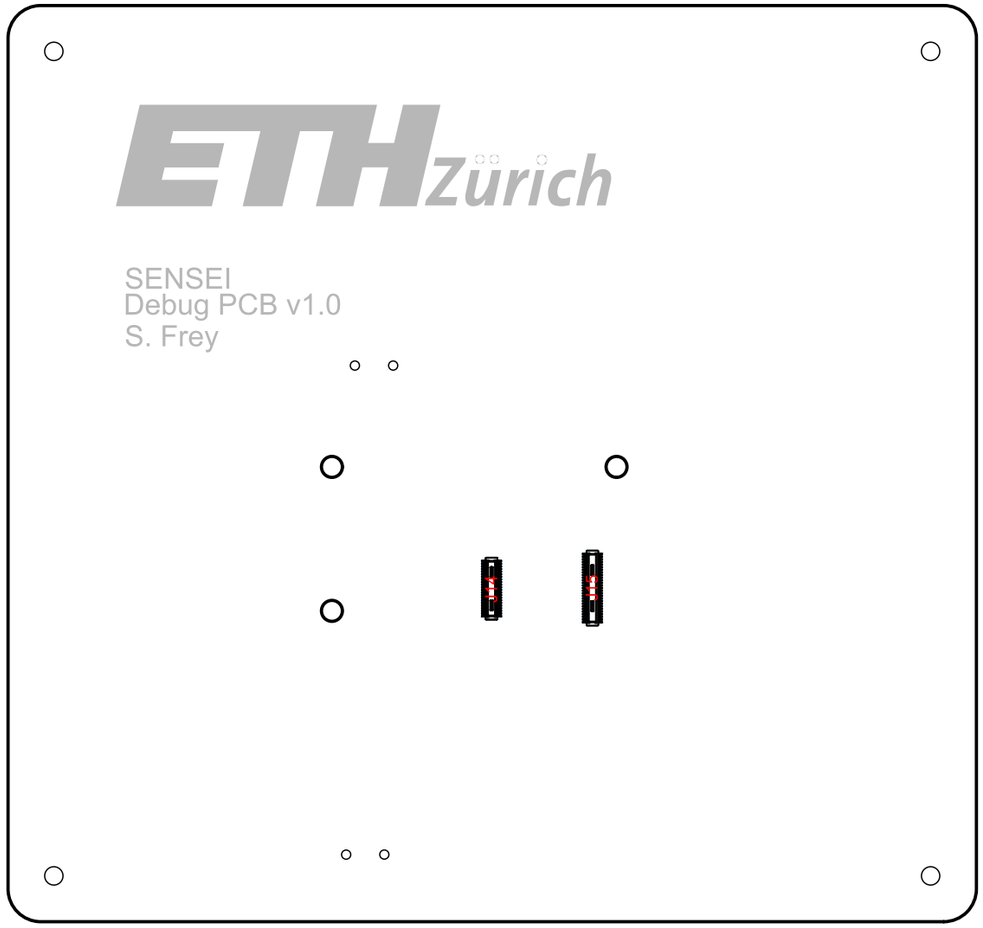
Drawing number: 1	Rev: v1.1	Format: A2	Laboratory: Integrated Systems Laboratory
Date: 18.12.2025 15:48:24		Drawn by: Sebastian Rey	Project: 01_gap_interface.SchDoc
File: C:\Users\is-fre\Github\pulp-bio\sensei-debug-board\Hardware\SENSEI_Debug-Board\01_gap_interface.SchDoc			Page: 1 of 1


SENSEI
Debug Board v1.1
S. Frey
P. Wiese
ETH zürich
MSP Debug
nRF Debug
GAP Debug
Reset
RGB
Green
Yellow
Red
GAP Boot select
eMRAM
JTAG
eFUSE
For SENSEI Base Board v1.0, make sure that the GAP9 is connected to the SPI bus (controlled by nRF)
Button 1
Button 2
Button 3
LED RGB
Connect jumpers to use buttons and LEDs
GND GND
J1
J2
J3
J4
J7
J8
J9
J10
J11
J12
J13
SW4
TP1
TP2
TP3
TP4

View from Top side (Scale 1:1)



View from Bottom side (Scale 1:1)



Eidgenössische Technische Hochschule Zürich Swiss Federal Institute of Technology Zurich	Project: **BioGAP ExG shield**		
Drawing number: =documentnumber Rev: v1.0 Date: 18.12.2025 15:48	Format: **A3**	Laboratory: * Drawn by: *	Sheet: DOC_ASSEMBLY.PCBDwf Page =sheetnumber of =sheettotal
File: C:\Users\s-fre\Github\pulp-bio\sensei-debug-board\Hardware\SENSEI_Debug-Board\DOC_ASSEMBLY.PCBDwf			

Comment	Description	Designator	Quantity	Resistance	Capacitance	Tolerance
CL05A105KP5NNNC	Cap Ceramic 1uF 10V X5R ±10% SMD 0402 +85°C Paper T/R	C1	1		1uF	10%
CL05A104KA5NNNC	Cap Ceramic 100nF 25V X5R ±10% SMD 0402 +85°C Paper T/R	C2	1		0.1uF	10%
SML-LX0404SIUPGUSB	SML-LX0404SIUPGUSB RGB LED	D1	1			
LTST-C170GKT	LED GREEN CLEAR SMD	D2	1			
LTST-C170YKT	LED YELLOW CLEAR SMD	D3	1			
LTST-C170EKT	LED RED CLEAR SMD	D4	1			
NCR421UX	Integrated Circuit	IC1, IC2, IC3	3			
3020-20-0300-00	Connector	J1, J2, J8, J9, J10	5			
DF40HC(3.0)-40DS-0.4V(51)	40 Position 0.4mm Pitch Dual Row Vertical Surface Mount (SMT) Pin Socket, 3.0mm Stacking Height	J3	1			
DF40HC(3.0)-50DS-0.4V(51)	50 Position 0.4mm Pitch Dual Row Vertical Surface Mount (SMT) Pin Socket, 3.0mm Stacking Height	J4	1			
TSM-110-01-T-DV	0.025" SQ Post Header, Surface Mount, Vertical, -55 to 105 degC, 2.54 mm Pitch, 20-Pin, Male, RoHS	J7	1			
3220-10-0300-00-TR	Connector	J11, J12, J13	3			
DF40C-40DP-0.4V(51)	50 Position 0.4mm Pitch Dual Row Vertical Surface Mount (SMT) Receptacle	J14	1			
DF40C-50DP-0.4V(51)	50 Position 0.4mm Pitch Dual Row Vertical Surface Mount (SMT) Receptacle,	J15	1			
RR1220Q-680-D	68Ω ±0.5% 0.1W 1/10W Chip Resistor 0805 (2012 Metric) Thin Film	R1, R2, R3	3	68R		0.5%
CRCW0201220RFKED		R4, R5, R6, R8, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20	15			
RK73B1JTTD562J		R7, R9, R21	3	5.6k		
PTS815 SJM 250 SMTR LFS	No Description Available	SW1, SW2, SW3, SW5, SW6, SW7	6			
JS203011JCQN	SWITCH SLIDE DP3T 300MA 6V	SW4	1			
TP THT		TP1, TP2, TP3, TP4	4			
IS31FL3194-CLS2-TR	LED Driver with Programmable Pattern Sequencing 3-Channel 40mA 8-Pin WCSP T/R	U1	1			